

Week-1 ML Ramp Up Plan

Group-wise Topic Distribution

Group	Design	Students
Group A	Titanic Dataset	4
Group B	Iris Dataset	4
Group C	UCI Heart Disease Dataset	4

Work Expectation per Group

Step	Activity	Description	Output
1	Learning	Understand concepts deeply	Learning Notes
2	Exploration	Apply on dataset	Experiment Logs
3	Implementation	Build ML workflow	Scripts / Notebooks
4	Output Generation	Structured outputs	JSON Files
5	Version Control	Track progress	Git Commits
6	Issue Analysis	Document problems	Issue Log

Presentation Guidelines & Structure Examples

❖ What Each Group Must Present

- Explain Dataset and Problem
- Explain pre-processing and feature engineering
- Explain model selection and behavior
- Show git repository structure
- Demonstrate JSON outputs
- Explain Approach and workflow
- Highlight Challenges and fixes
- Show final pipeline and result

❖ Presentation Expectations

- Keep explanations simple and logical
- Focus on understanding, not memorization
- Use examples and snippets for your own work
- Be ready to answer questions from any part of your topic

- Every member should be able to explain the group work

❖ Examples of What You Will Be Asked

- Why did you design it this way?
- What was the reason for choosing this instead of something else?
- What happens if this fails?
- What assumptions did you make?
- What part was most difficult?

❖ Output Requirement

All results must be stored in structured JSON format, including:

- Dataset summary
- EDA statistics
- Feature transformations
- Model predictions
- Performance metrics

❖ Example of JSON Output Structure

```
{
  "dataset": "Iris",
  "features": ["sepal_length", "sepal_width"],
  "model": "LogisticRegression",
  "metrics": {
    "accuracy": 0.95,
    "precision": 0.94,
    "recall": 0.93
  }
}
```

❖ Example of Git Repository Structure

```
ml-ramp-up-week1/
|
├─ data/
|   └─ dataset.csv
|
├─ notebooks/
|   └─ dayX_analysis.ipynb
|
├─ src/
|   └─ scripts (processing / training)
|
├─ outputs/
|   ├── day1_data.json
|   ├── day2_eda.json
|   ├── day3_features.json
|   ├── day4_models.json
|   ├── day5_eval.json
|   └─ day6_pipeline.json
|
├─ models/
|   └─ pipeline.pkl
|
├─ logs/
|   └─ issue_log.txt
|
└─ README.md
```

Rules & Policy (Week-1 RTL Assignment)

❖ Core Rules

- Each group works independently
- No cross-group dependency or comparison
- Focus on deep understanding, not covering everything

❖ Work Rules

- All members must understand assigned topics
- Tasks can be divided, but knowledge must be shared internally
- Maintain daily learning and experiment notes
- Produce real outputs (even if incomplete)

❖ Documentation Rules

- Maintain:
 - Learning notes
 - Experiment logs
 - Outputs
 - Issue log
- Clearly document:
 - What worked
 - What failed
 - What is not understood

❖ Issue Logging Rules (Very Important)

- Be honest about problems
- Clearly mention:
 - Errors
 - Confusions
 - Assumptions
- Do NOT hide mistakes

❖ Behavior Rules

- Do not copy without understanding
- Do not depend on other groups
- Try solving before asking
- Discuss within your group

❖ Evaluation Focus: You will be evaluated based on:

- Depth of understanding
- Clarity of explanation
- Effort and approach
- Issue identification
- Ability to think and explain

Topics Exploration and Experimental Work

❖ Day 1: Data Understanding & Repository Setup

Basic Terms (Understand First)

- Dataset
- Features
- Label
- DataFrame
- JSON

Learning Topics

- numpy basics
- pandas (read_csv, filtering, groupby)
- dataset structure understanding

Experimental work

- Step 1: Create git repositories
 - Connect your local git to github using ssh
 - Push your first few files there and make basic folder structure
- Step 2: Find out the dataset from Kaggle and download
- Step 3: Load dataset using pandas in your notebook/python file
- Step 4: Identify various aspects from dataset
- Step 5: Validate Extraction
 - Features
 - Targets
- Step 6: Generate Dataset Summary
 - Name as yourname_week1day1_data.json

❖ Day 2: Exploratory Data Analysis & JSON Structuring

Basic Terms (Understand First)

- Mean
- Std
- Distribution
- Correlation

Learning Topics

- Descriptive Statistics

- Histogram, Boxplots
- correlation heatmap
- class imbalance

Experimental work

- Step 1: Perform EDA
- Step 2: Generate stats and observations
- Step 3: Generate Plots and store them appropriately
- Step 4: Convert findings into structured JSON, follow similar naming conventions

❖ Day 3: Feature Engineering

Basic Terms (Understand First)

- Normalization
- Encoding
- Feature Understanding

Learning Topics

- Various Scaling Techniques
- Various data preprocessing steps and their importance
- Different data type of feature and their encoding techniques
- Importance of feature engineering and their techniques and creating new features

Experimental Work

- Step 1: Do data preprocessing tasks and clean data
- Step 2: Apply various transformations to data
- Step 3: Create new features
- Step 4: Document using similar naming conventions

❖ Day 4: Model Training and Prediction Logging

Basic Terms (Understand First)

- Machine Learning Models
- Training the models
- Inference
- Hyperparameter

Learning Topics

- What are machine learning models?

- What are the different kinds of models are there?
- Why do we need different machine learning models for different datasets?
- How split for train test and why is it needed.

Experimental Work

- Step 1: Choose your model and understand why
- Step 2: Split dataset, various ways to do that, explain your choice
- Step 3: Train models, understand hyper parameters
- Step 4: Generate predictions
- Step 5: Store Predictions with appropriate naming convention

❖ Day 5: Evaluation & Performance Metrics

Basic Terms (Understand First)

- Precision
- Recall
- F1 score
- Accuracy

Learning Topics

- Confusion matrix & why is it important.
- Various evaluation metrics and their significance
- Data leakage and why it is important
- Why do we use cross validation?

Experimental Work

- Step 1: Compute Various Performance Metrics
- Step 2: Perform Cross Validation
- Step 3: Try to explain reasons behind the values
- Step 4: Store Metrics in json format with appropriate naming convention

❖ Day 6: Final Integration and Whole Pipeline Structure

Basic Terms (Understand First)

- Pipeline
- Model Export

Learning Topics

- How to create a pipeline for the whole flow from sklearn
- Train multiple models at once
- Trained models for reuse

Experimental Work

- Step 1: Use all your previous concepts and use at once
- Step 2: Train multiple models and analyze their outputs
- Step 3: Save the models for reuse on other datasets
- Step 4: Save Outputs of model evaluation metrics in JSON format with appropriate naming conventions
- Step 6: Identify Errors / Issues: Students must log:
 - Errors occurred
 - Any missing output values

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